Johnny Vogt
DS7330: Mini Project 3

a. How many copies of the book titled The Lost Tribe are owned by the library branch whose name is ‘Sharpstown’?
πLost_Tribe_Sharpstown(sum(σ(Title = “The Lost Tribe” and Branch_name = “Sharpstown”) (Book U Library_Branch ⋈ Book_Copies))

b. How many copies of the book titled The Lost Tribe are owned by each library branch?
πLost_Tribe_All (sum(σ(Title = “The Lost Tribe”) (Book ⋈ Book_Copies ⋈ Library_Branch))

c. Retrieve the names of all borrowers who do not have any books checked out.
πborrower_no_books(πCard_no_borrower(σ(Card_no = *)(Borrower)) – πCard_no_book_loan(σ(Card_no =*)(Book_loan)))

d. For each book that is loaned out from the Sharpstown branch and whose Due_date is today, retrieve the book title, the borrower’s name, and the borrower’s address.
πTitle_Name_Address(σ(Branch_Name = “Sharpstown” and Due_date = <Today>) (Book U Book_Loans ⋈ Library_Branch))

e. For each library branch, retrieve the branch name and the total number of books loaned out from that branch.
πBranch_Name_Loan(σ(Branch_Name) COUNT(σ(Book_ID))(Library_Branch ⋈ Book_Loan))

f. Retrieve the names, addresses, and number of books checked out for all borrowers who have more than five books checked out.
πName_Addr_BookLoan(σ(Name)σ(Address)σ(BookID(COUNT(Book_ID) > 5)) (Book_Loan ⋈ Borrower))

g. Retrieve a list of Book ID and Book Title that are borrowed by all borrowers.
πBookID_Title_All(σ(Book_ID)σ(Title) ((Book U Book_Loan) ⋈ Borrower))